

Personal information

Full name: Thibeau Wouters Email address: thibeauwouters@gmail.com
Location: Utrecht, Netherlands Website: thibeauwouters.github.io

Research interest

Gravitational waves & binary neutron star mergers Neutron star equation of state
Machine learning and differentiable programming GPU-accelerated Bayesian inference

Professional experience

2023 - Present **PhD Candidate in Astrophysics** Utrecht University
Advisor: Prof. dr. Chris Van Den Broeck
Topic: Bayesian inference on neutron star equation of state and gravitational waves

Education

2022 - 2023 **M. Sc. in Artificial Intelligence** KU Leuven
89.00% summa cum laude
Advisors: Prof. Dr. Tjonnje G. F. Li & Prof. Dr. Johan Suykens
2020 - 2022 **M.Sc. in Theoretical Physics** (minor: nuclear physics) KU Leuven
91.41% summa cum laude with congratulations of the Board of Examiners
Advisor: Prof. Dr. Nikolay Bobev
2017 - 2020 **B.Sc. in Physics** (74.42% cum laude) KU Leuven
2017 - 2020 **B.Sc. in Mathematics** (73.92% cum laude) KU Leuven

Publication and citation summary (from inspirehep.net on 8th September, 2026)

	Excluding LVK/ET [†]	all papers	Including LVK/ET collaboration papers	Including all LVK/ET papers
Number of papers:	19		40	59
Number of citations:	196		2,700	2,896
Average citations per paper:	10.3		67.5	49.1
h-index:	9		20	22

[†]LVK = LIGO-Virgo-KAGRA Collaboration, ET = Einstein Telescope Collaboration

Key publications (full list [here](#))

T. Wouters et al. (2025). *Phys. Rev. D* **112** (4), p. **043037**
– Development of GPU-accelerated software for solving the Tolman-Oppenheimer-Volkoff equations

T. Wouters et al. (2024). *Phys. Rev. D* **110** (8), p. **083033**
– Development of GPU-accelerated software for analyzing gravitational wave signals of binary neutron star mergers

D. Yallup et al. (2026). arXiv:2607.28265 [gr-qc]
– Real-time gravitational-wave parameter estimation without pre-trained neural networks, using flowMC/jim-based ab initio inference

H. Koehn et al. (2026). arXiv:2607.28438 [astro-ph.HE]
– Multi-messenger detection prospects and constraints on the neutron star equation of state, mass distribution, and cosmology for next-generation gravitational-wave detectors

T. Wouters et al. (2026). *Phys. Rev. D* **114** (4), p. **043005**
– Physics-informed neural-flow priors that incorporate neutron star equation-of-state knowledge into gravitational-wave inference

Prizes

2022 KU Leuven Physics prize: highest grade among physics master students

Teaching record

Student supervision

PhD

2026 M. Sc. data science

Master

2026 Frank Snijder M. Sc. data science

2025 - 2026 Marlinde Drent M. Sc. data science

2025 Oleksii Bevza M. Sc. data science

2025 Hong Ye Guo M. Sc. data science

Bachelor

2025 Elin Jakobs B. Sc. data science

2025 Senna van Os B. Sc. physics

2024 Frank Snijder B. Sc. physics

2024 Tom Oeloff B. Sc. physics

2024 Marlinde Drent B. Sc. physics

Courses tutored

2023 Bachelor Gravitational waves Utrecht University

2024 Master Gravitational waves Utrecht University

2023 Bachelor Optics and astronomy Utrecht University

2020 - 2022 Bachelor Introductory mathematics and physics courses KU Leuven

2020 Master Quantum mechanics (Peer Assisted Learning tutor) KU Leuven

2019 - 2023 All levels Private tutoring, Student / Slaagsleutels / Spring-Stof –

Invited summer school teachings

2020 - 2022 Summer schools, KU Leuven

Public and professional services

Leadership roles

2022 - 2023 Student representative responsible for the quality of education of the AI master program, KU Leuven

Invited talks

2025 (Sep) Seminar AEI Potsdam, Germany

2025 (Oct) 1st Symposium on Accelerated Computing for Application Workflows SURF, Netherlands

2025 (Nov) Advanced Computing User Day SURF, Netherlands

Conference talks

2023 ML4GW Workshop, Utrecht

2025 ET Symposium

2025 EuCAIFCon

2025 AIlands

2025 NNV Meeting (Dutch Physics Society)

2026 GWfreeride (flash talk)

Public engagement

• Volunteer at the Einstein Telescope stand, Nerdland (2025): the largest popular science festival in Belgium
